



High Tide Low Tide — FULL MIND MAP

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Chapter 24_High_Tide_Low_Tide

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Tides — Formation ka Karan

- High-Low tide formation: **Gravitation + Centripetal force + Centrifugal force** (Earth-Moon gravitational pull + inhi 2 forces ka combined effect). [Q1,Q2]
- Moon-facing side pe **direct high tide** (gravity dominant); opposite side pe **indirect high tide** (Earth ka centrifugal force ki wajah se). [Q3]
- Earth ~24 hours 50 minutes mein **2 high tides aur 2 low tides** experience karta hai. [Q1]

Neap Tide

- Neap tide ke time: high tide normal se ~20% **kam** hota hai, low tide normal se **zyada** hota hai (Statement A **TRUE**); Neap tide **new-moon/full-moon PE NAHI** hoti — inke ~7th/8th din hoti hai jab Sun-Moon Earth ke sapeksh ~right angle pe hote hain (Reason R **FALSE**). [Q4]



EXAM TRAP

- Neap tide new-moon/full-moon pe NAHI hoti — yeh common confusion hai (Spring tide in dono din hoti hai, Neap tide unke beech ~7-8 din baad). [Q4]

Spring Tide

- Spring Tide **Sun-Earth-Moon straight line (syzygy)** mein hone par hoti hai — dono Full Moon aur New Moon positions (M2 aur M4) spring-tide conditions represent karti hain. [Q5,Q6]

High Tide — Main Cause

- Ocean mein high tide mainly **Moon** ki wajah se hoti hai (Moon Earth ke bahut close hai, isliye Sun se zyada mass hone ke bawajood Moon ka tide-effect zyada hota hai — Moon ka influence Sun se ~2x+ zyada hai). [Q7]